AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 13 · STUDENT LESSON

Effect size and practical importance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Define the gain. State the scale. Apply the practical rule.

Report paired-time and error-count effects without confusing relative change, mean criteria and statistical uncertainty.

Keep one habit: Effect contract → paired-gain ledger → native error-threshold picture → sign/practicality probes → complete descriptive audit → independent smaller gain.

MAIN LESSON · SECTION 1

Report how much changed and whether it meets the stated practical criterion

> A positive difference, practical importance and statistical evidence are different judgments.

Learning goals

I can explain the **mechanism** and terminology of **effect size** and practical importance.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve interval limits, comparison direction and task constraints

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Uncertainty intervals. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain **effect size** and practical importance using the records and boundaries of this fictional investigation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **effect size** and practical importance

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal
</details>

MAIN LESSON · SECTION 2

Define the effect scale, direction and prespecified threshold

> An observed magnitude is useful to report even when it misses the criterion.

Begin with “faster by enough?”

A candidate can be faster without being faster by enough for the task. Define the effect on a clear scale, then compare it with a minimum useful change declared before inspecting results. Here gain means baseline time minus candidate time, so positive means the candidate is faster. That sign convention differs from the previous lesson’s B–A time difference.

Keep the units, pairing and threshold visible rather than relying on the word “better”.

Define effect size and practical importance

Effect size describes magnitude on a stated scale. Practical importance depends on task requirements, not merely a nonzero difference.

Complete paired-gain and separate error-rule contract

Effect/decision item	Complete supplied-exercise rule
Main paired cases	Q1: 10→8; Q2: 12→9; Q3: 8→7 seconds, baseline→candidate
Gain definition	Baseline seconds – candidate seconds; positive is faster candidate
Summary	Mean of the three paired gains, in seconds
Prespecified useful-gain rule	Mean gain at least 2 seconds; equality meets the rule
Mean versus individual cases	Rule applies to the mean, not a requirement that every case gains 2 seconds
Separate error-count picture	A 20 errors/1000 tasks; B 15/1000 initially, changed B 8/1000
Picture useful-change rule	At least 10 fewer errors per 1000; fictional task criterion, not a universal threshold
Absolute and relative effects	Keep errors-per 1000 and percentage points separate from relative reduction

Effect/decision item	Complete supplied-exercise rule
Evidence limits	No uncertainty, significance, causal effect or complete deployment-readiness evidence supplied
Other task constraints	Quality, resources and robustness require their own observations; passing one criterion does not settle them

Name supplied and missing evidence

> **List the given facts and one fact that is not supplied.**

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace paired gains before averaging or deciding practical value

> The average meets its rule without every individual case doing so.

Original paired-gain calculation

For paired response durations compute baseline minus candidate per case and average the differences. Positive means faster candidate. Compare with a prespecified minimum useful gain.

Complete case-by-case arithmetic and mean decision

Main paired case	Baseline seconds	Candidate seconds	Gain: baseline – candidate
Q1	10	8	2
Q2	12	9	3
Q3	8	7	1
Mean gain	(2+3+1)/3	2 seconds	Meets inclusive mean threshold 2
Individual variation	Q3 gain is1 second	Mean rule still met	Not a claim every case meets an individual 2-second target
Remaining uncertainty	Only three supplied paired cases	No interval or statistical test supplied	No population-certainty or causal claim

State the threshold and missing broader evidence

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Effect size and practical importance — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Baseline times 10, 12, 8; candidate 8, 9, 7 seconds. Useful mean gain requires at least 2 seconds.	For paired response durations compute baseline minus candidate per case and average the differences. Positive means faster candidate. Compare with a prespecified minimum useful gain.	Paired gains 2, 3, 1; mean 2 meets the inclusive useful-gain threshold. It is a three-case descriptive result, not a population certainty.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict a larger error-count reduction under the same practical rule

> A new result may meet the threshold without adding uncertainty evidence.

Current task and response

- Baseline times 10, 12, 8; candidate 8, 9, 7 seconds. Useful mean gain requires at least 2 seconds. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict changed B count of eight errors

- Change: B makes 8 errors in 1000.

▼ Check after predicting

Prediction: The reduction becomes 12 per 1000 and meets the stated **practical threshold**.

Reason: Twenty minus eight exceeds the required ten fewer errors.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read absolute effect, relative change and practical rule together

> Keep error-count units separate from the main paired time gains.

Main paired-time criterion and checked result

Baseline times 10, 12, 8; candidate 8, 9, 7 seconds. Useful mean gain requires at least 2 seconds.

Paired gains 2, 3, 1; mean 2 meets the inclusive useful-gain threshold. It is a three-case descriptive result, not a population certainty.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate error-count inputs and magnitude definitions

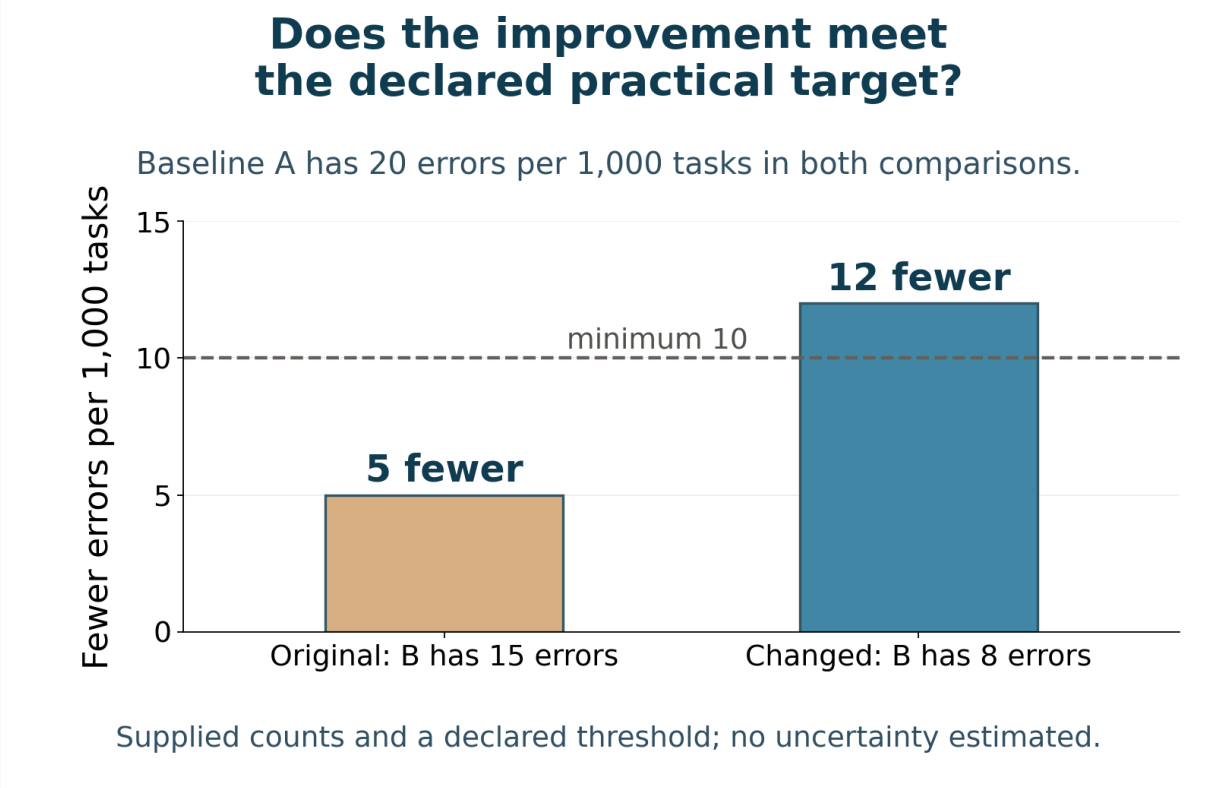
An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**.

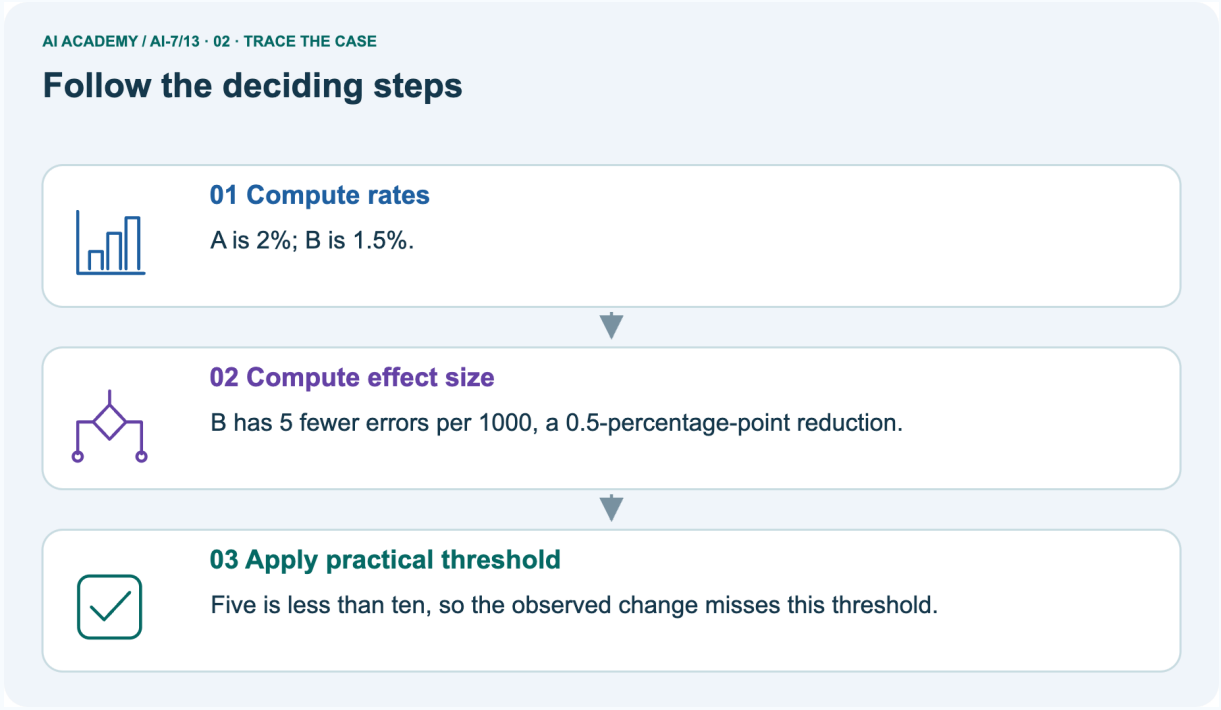
A makes 20 errors in 1000 tasks; B makes 15. Practical improvement requires at least 10 fewer errors per 1000.

The picture is a separate error-rate comparison on one thousand tasks per version. A has 20 errors, or2%; B has 15, or1.5%. The reduction is five errors per thousand, equal to0.5 percentage points and 25% relative to A's error count. The declared practical rule requires at least ten fewer errors per thousand, so this observed reduction misses it.

Changed B with eight errors gives twelve fewer, a1.2-percentage-point reduction, meeting that rule. These descriptive counts supply no uncertainty interval or significance result. Meeting this one criterion is not automatic evidence of all quality, resource or deployment requirements.

Compare original/changed effects in the native threshold chart





Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute rates	A is 2%; B is 1.5%.
2. Compute effect size	B has 5 fewer errors per 1000, a 0.5-percentage-point reduction.
3. Apply practical threshold	Five is less than ten, so the observed change misses this threshold.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Effect size describes how much the outcome changes. Practical importance depends on the real decision criterion; relative changes can sound large while absolute changes remain small.

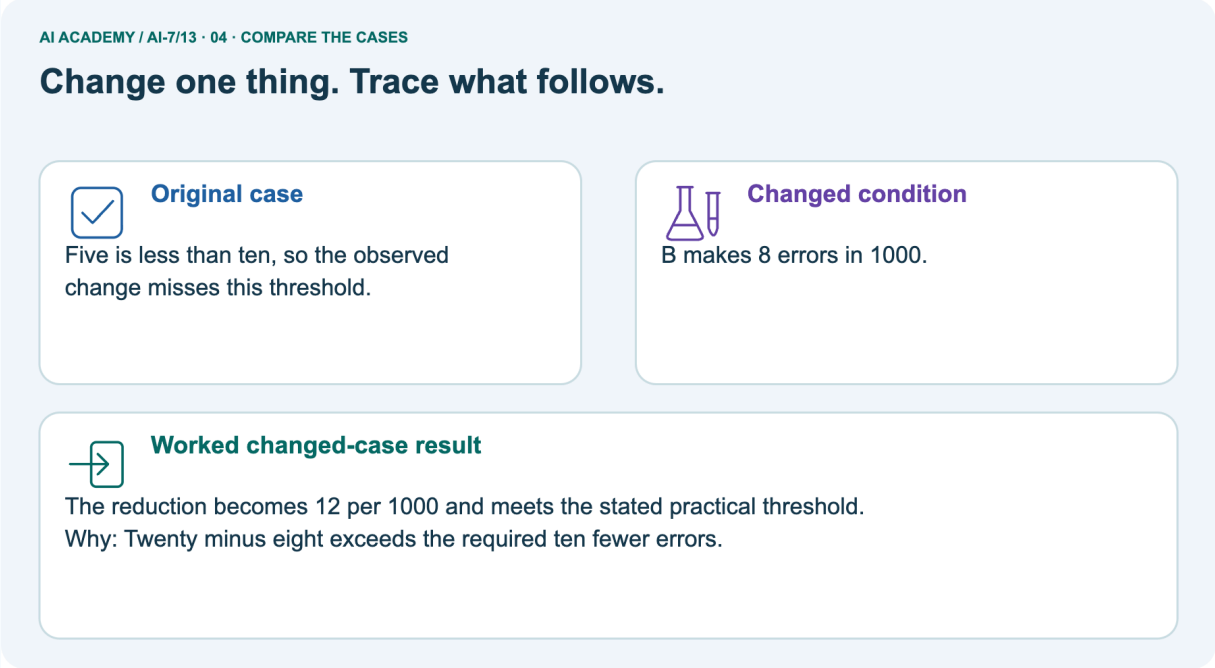
Complete counts, rates and practical decisions

Separate error comparison	Original B	Changed B
A baseline error count	20/1000 =2%	Same20/1000 =2%
B error count/rate	15/1000 =1.5%	8/1000 =0.8%
Fewer errors per 1000	20−15 =5	20−8 =12
Absolute rate reduction	0.5 percentage points	1.2 percentage points
Relative error reduction	5/20 =25%	12/20 =60%
At least 10 fewer required	Misses:5 <10	Meets:12 ≥10
Statistical uncertainty	Not supplied	Still not supplied

Predict B with eight errors

- Change one thing: B makes 8 errors in 1000. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain missing uncertainty and absolute-effect interpretation

- No uncertainty or statistical significance is supplied; these counts alone cannot establish a general effect.
- Read the labelled picture: Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?

▼ Earlier wording, alternatives and picture transcripts

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def paired_gain(baseline, candidate):
    if not baseline or len(baseline) != len(candidate):
        raise ValueError('paired lengths')
```

```
    differences = [a - b for (a, b) in zip(baseline, candidate)]
    return (differences, sum(differences) / len(differences))

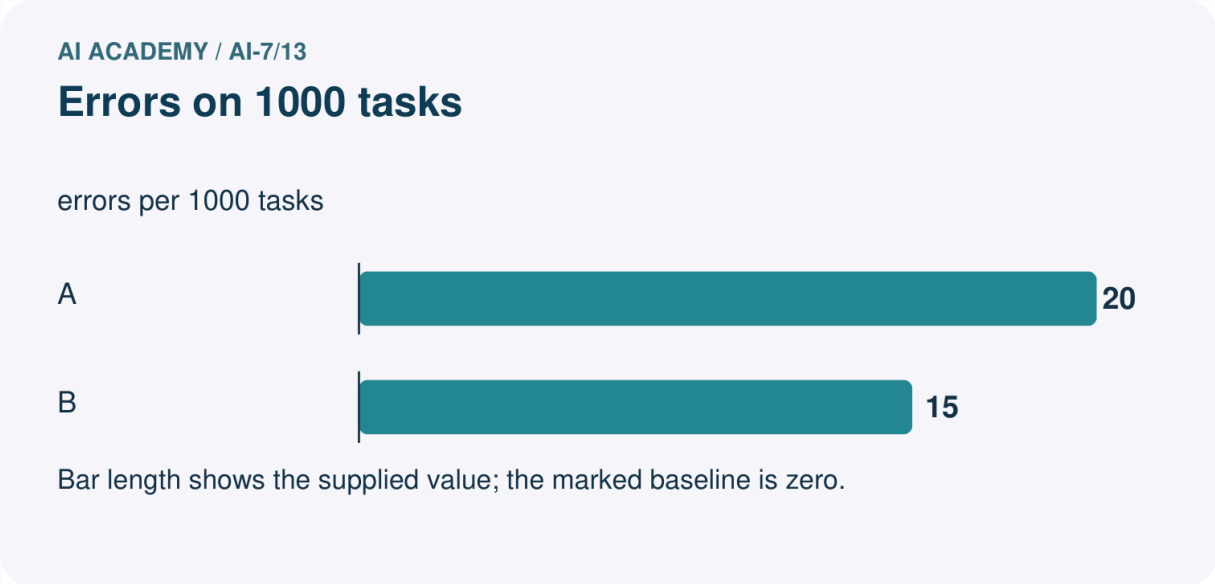
print(paired_gain([10, 12, 8], [8, 9, 7]))
```

Learn from the worked case

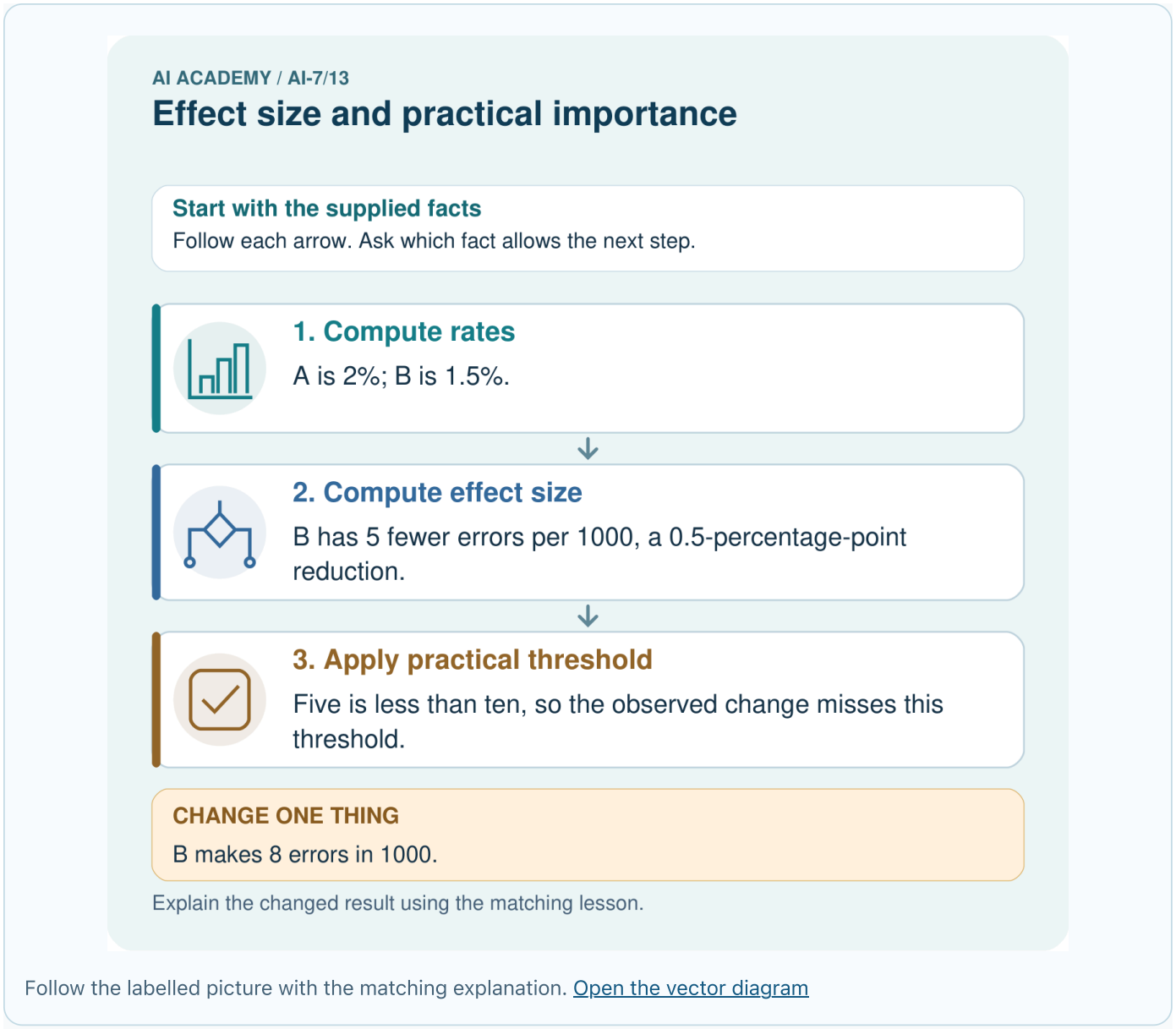
Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Effect size and practical importance

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Audit sign conventions and the reach of a practical decision

> Magnitude, decision criteria and statistical uncertainty need distinct evidence.

Core effect-size qualification and failure analysis

A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete mean/individual and absolute/relative probes

Effect/practicality probe	Expected interpretation
Positive baseline-minus-candidate gain	Candidate is faster under this sign convention
Mean equals the minimum useful gain	Meets an at-least threshold
One case below mean target	Does not automatically fail a rule about the mean
Relative reduction sounds large	Report absolute units and baseline rate too
Observed reduction below practical target	Still a descriptive change, but criterion not met
Observed criterion met	No automatic significance, causal or deployment conclusion
Threshold chosen after inspecting outcomes	Disclose revision rather than claim prespecified importance
Faster response is wrong or over resource budget	Other requirements can still block practical usefulness
Three supplied pairs	Do not invent a population effect interval or certainty

Guided repair and explanation

- A report omits this qualification: A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise naming the effect and comparing it with the right rule

> Write both the observed magnitude and the criterion outcome.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported W2 trace

Use workbook W2 for the connected example "**Effect size** and practical importance". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for **Effect size** and practical importance. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?

My explanation using a pictured step and the given facts

Reserve the selected new paired times and threshold

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run the complete descriptive-effect and practical-criterion investigation

> Inspect the paired records, units and exact decision rule before interpreting the flags.

Current paired-time gain investigation

Baseline times 10, 12, 8; candidate 8, 9, 7 seconds. Useful mean gain requires at least 2 seconds. Show the intermediate steps and give the conclusion with its scope.

Runnable complete time and count analysis

```
"""Supplied paired gains and practical thresholds, without inference or live timing."""
from math import isfinite
from statistics import mean
from fractions import Fraction

def finite_nonnegative(value):
 return type(value) in (int,float) and isfinite(value) and value>=0

def paired_gain(rows,minimum_useful_gain):
 if type(rows) is not list or not rows or not finite_nonnegative(minimum_useful_gain) or
minimum_useful_gain<=0:
  raise ValueError('paired records and positive declared threshold required')
 ids=[];gains=[]
 for row in rows:
  if type(row) is not dict or set(row)!={'case_id','baseline_seconds','candidate_seconds'}:
   raise ValueError('exact paired case/timing fields required')
  if type(row['case_id']) is not str or not row['case_id'] or row['case_id'] in ids:raise
ValueError('unique paired case IDs required')
  if not all(finite_nonnegative(row[k]) for k in ['baseline_seconds','candidate_seconds']):raise
```

```
ValueError('finite nonnegative supplied times required')
  ids.append(row['case_id']);gains.append(row['baseline_seconds']-row['candidate_seconds'])
 average=mean(gains)
 return {'case_ids':ids,'gains_seconds':gains,'mean_gain_seconds':average,

'minimum_useful_mean_gain':minimum_useful_gain,'meets_declared_mean_threshold':average>=minimum_useful_gain}
def error_reduction(a_errors,b_errors,total,minimum_per_thousand=10):
 if any(type(v) is not int for v in [a_errors,b_errors,total,minimum_per_thousand]) or total<=0 or
minimum_per_thousand<=0:
  raise ValueError('exact integer counts and positive denominator/threshold required')
 if not 0<=a_errors<=total or not 0<=b_errors<=total:raise ValueError('valid error counts required')
 difference=Fraction(a_errors-b_errors,total)
 return {'A_error_rate':Fraction(a_errors,total),'B_error_rate':Fraction(b_errors,total),
         'reduction_percentage_points':100*difference,'fewer_errors_per_thousand':1000*difference,
         'relative_error_reduction':Fraction(a_errors-b_errors,a_errors) if a_errors else None,
         'meets_declared_threshold':1000*difference>=minimum_per_thousand}
rows=[{'case_id':'Q'+str(i+1),'baseline_seconds':a,'candidate_seconds':b}
      for i,(a,b) in enumerate(zip([10,12,8],[8,9,7]))]
main=paired_gain(rows,2)
assert main['gains_seconds']==[2,3,1] and main['mean_gain_seconds']==2 and
main['meets_declared_mean_threshold']
picture=error_reduction(20,15,1000)
changed=error_reduction(20,8,1000)
assert picture['fewer_errors_per_thousand']==5 and picture['reduction_percentage_points']==Fraction(1,2)
and not picture['meets_declared_threshold']
assert picture['relative_error_reduction']==Fraction(1,4)
assert changed['fewer_errors_per_thousand']==12 and changed['reduction_percentage_points']==Fraction(6,5)
and changed['meets_declared_threshold']
assert error_reduction(20,10,1000)['meets_declared_threshold']
print('Main paired baseline-minus-candidate gains:',main)
print('Separate original supplied error-count comparison:',picture)
print('Changed supplied B error count:',changed)
print('Main mean threshold is inclusive and applies to the mean, not every case. The prior lesson used B-
A; this gain definition deliberately uses A-B.')
print('These are descriptive supplied observations and fictional practical criteria, not actual timings,
significance, uncertainty intervals, causal effects or complete deployment decisions. Quality/resources
require their own evidence.')
```

Expected magnitudes and evidence limits

1. Predict the three gains, their mean and equality with the two-second mean threshold. The complete code keeps paired IDs, finite nonnegative supplied times and a declared positive threshold. In the separate count comparison it computes exact absolute/relative reductions, checks the ten-per-thousand criterion and includes the equality boundary. None of the arithmetic times a real response or estimates uncertainty.

   A passing mean criterion does not mean every case gains two seconds, and a relative reduction is not the percentage-point change. Keep the independent three-case times and 1.5-second threshold for the first individual attempt.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

```
def paired_gain(baseline, candidate):
    if not baseline or len(baseline) != len(candidate):
        raise ValueError('paired lengths')
    differences = [a - b for (a, b) in zip(baseline, candidate)]
    return (differences, sum(differences) / len(differences))

print(paired_gain([10, 12, 8], [8, 9, 7]))
```

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: a positive gain that may be too small

> Calculate the paired mean and apply the new declared threshold.

Complete selected independent task

Baseline times are 6, 8, 10 seconds; candidate times are 5, 7, 9 seconds on the same three cases in the same order. Define each gain as baseline minus candidate. Useful mean gain must be at least 1.5 seconds. Calculate and decide.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Baseline times are 6, 8, 10 seconds; candidate times are 5, 7, 9 seconds on the same three cases in the same order. Define each gain as baseline minus candidate. Useful mean gain must be at least 1.5 seconds. Calculate and decide.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the effect definition, threshold and remaining task constraints

> Preserve improvements and failures without turning either into unsupported general conclusions.

Project task and artifact

Add a labelled **effect size** and practical importance **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **effect size** and practical importance.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local calculation or paper trace and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ Earlier wording, alternatives and picture transcripts

Evidence for your project

MAIN LESSON · SECTION 16

Explain the magnitude, practical decision and unknown uncertainty separately

> A useful report keeps all three visible.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **effect size** and practical importance

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Effect size	The magnitude of a difference under a declared scale
Paired difference	A within-unit comparison
Practical threshold	A prespecified magnitude considered useful

Use two terms to explain a step in this week's example.

▼ **Extra challenge · Explain a stronger case**

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ **Reference · Check the research reasoning and its limits**

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Effect size and practical importance — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Baseline times 10, 12, 8; candidate 8, 9, 7 seconds. Useful mean gain requires at least 2 seconds.	For paired response durations compute baseline minus candidate per case and average the differences. Positive means faster candidate. Compare with a prespecified minimum useful gain.	Paired gains 2, 3, 1; mean 2 meets the inclusive useful-gain threshold. It is a three-case descriptive result, not a population certainty.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**.

By the end, I can	How I will show it
Explain the mechanism and terminology of effect size and practical importance	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**.

- Which labelled step matches this detail? Five is less than ten, so the observed change misses this threshold.
- 1. Apply **practical threshold**
- 2. Compute rates
- 3. Compute **effect size**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Apply **practical threshold** Five is less than ten, so the observed change misses this threshold. **Effect size** describes how much the outcome changes.

Practical importance depends on the real decision criterion; relative changes can sound large while absolute changes remain small.

Changed-case question and optional hint

- Changed-case practice: B makes 8 errors in 1000. Which conclusion is supported by the supplied facts?
- 1. The reduction becomes 12 per 1000 and meets the stated **practical threshold**.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The reduction becomes 12 per 1000 and meets the stated **practical threshold**. Twenty minus eight exceeds the required ten fewer errors.

Repair and retain the record

Review Uncertainty intervals (ai-7/12). Goal: Explain the **mechanism** and terminology of uncertainty intervals. Short practice: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2].

Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].. Return to this week and explain what you changed.

Review Confounders and controls (ai-7/10). Goal: Explain the **mechanism** and terminology of confounders and controls. Short practice: Draw the four condition/background combinations.

Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?. Return to this week and explain what you changed.

Review Hypotheses and measurable outcomes (ai-7/7). Goal: Explain the **mechanism** and terminology of hypotheses and measurable outcomes. Short practice: Name the predicted outcome and how unsupported is scored.

If B sounds friendlier but ties A's unsupported count, does the recorded improvement hypothesis receive support?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

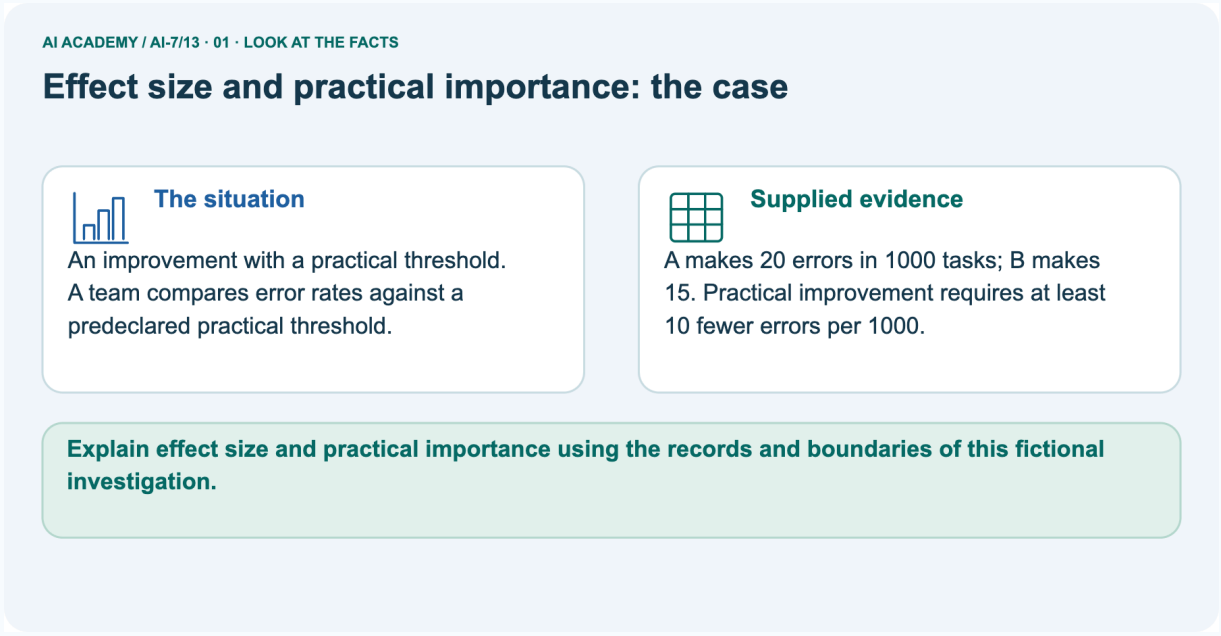
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Effect size and practical importance: the case



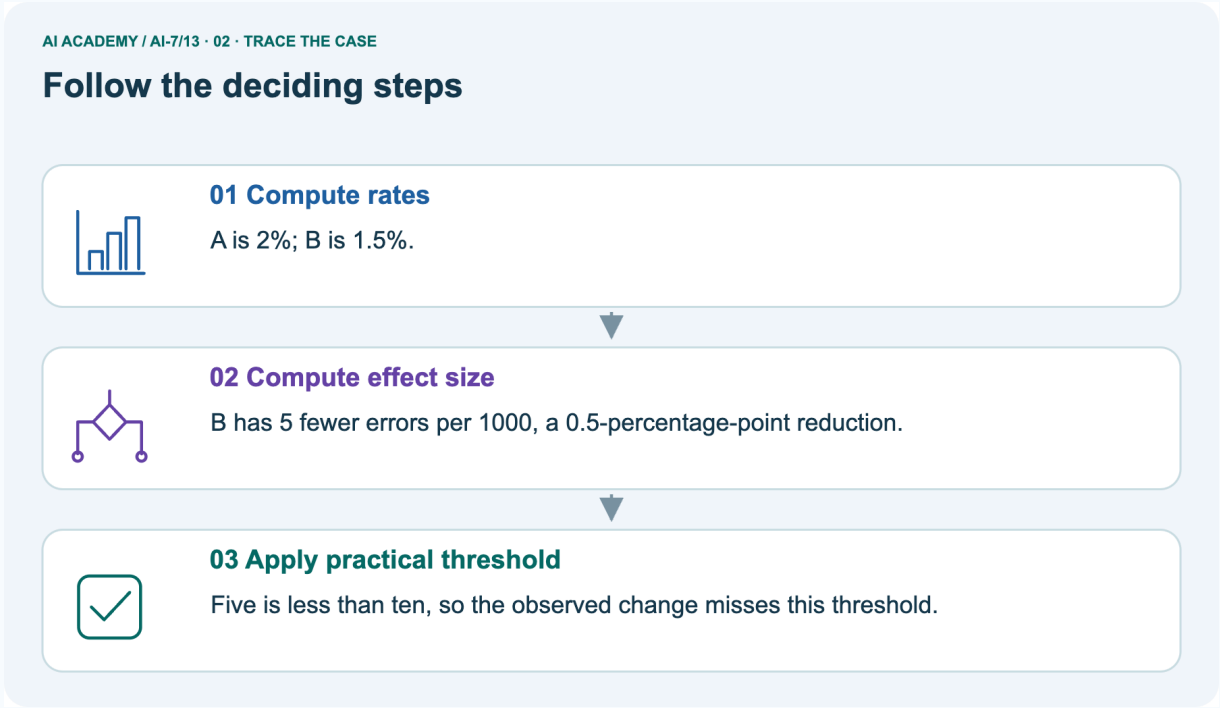
Follow the labelled picture with the matching explanation. Open the vector diagram

An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**. Supplied facts: A makes 20 errors in 1000 tasks; B makes 15.

Practical improvement requires at least 10 fewer errors per 1000.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Compute rates: A is 2%; B is 1.5%. → Compute **effect size**: B has 5 fewer errors per 1000, a 0.5-percentage-point reduction. → Apply **practical threshold**: Five is less than ten, so the observed change misses this threshold.

Point to each numbered stage and explain which recorded **input** it uses.

03. Errors on 1000 tasks

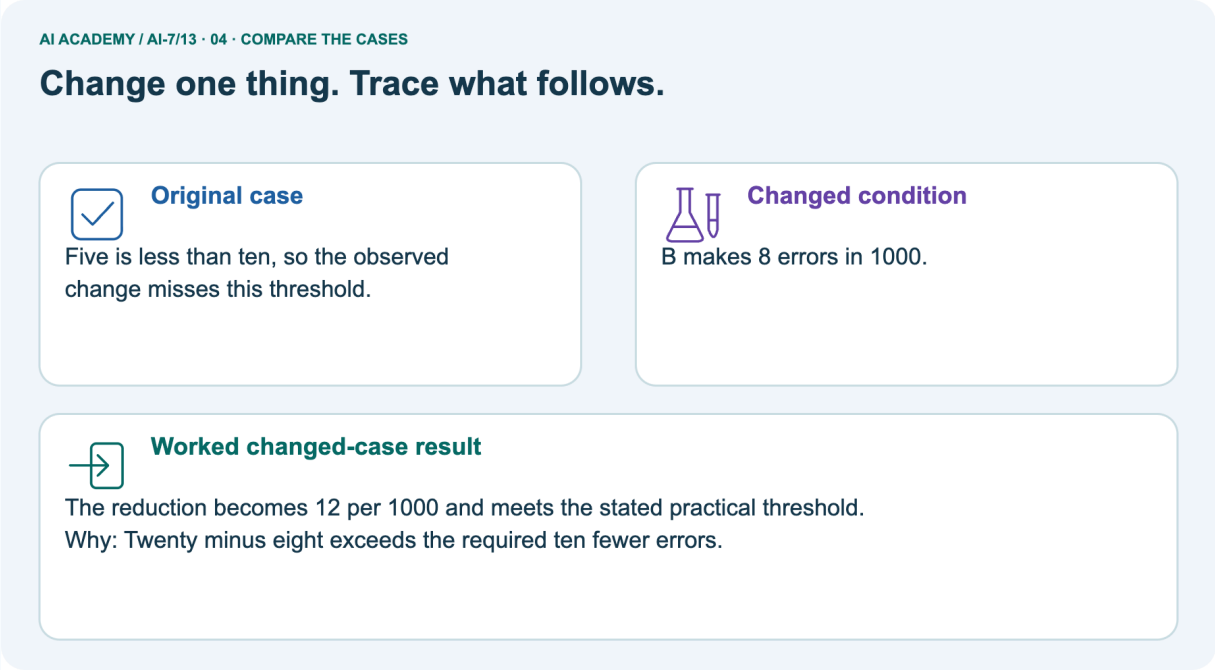


Follow the labelled picture with the matching explanation. Open the vector diagram

Errors on 1000 tasks. Units: errors per 1000 tasks. A: 20. B: 15.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



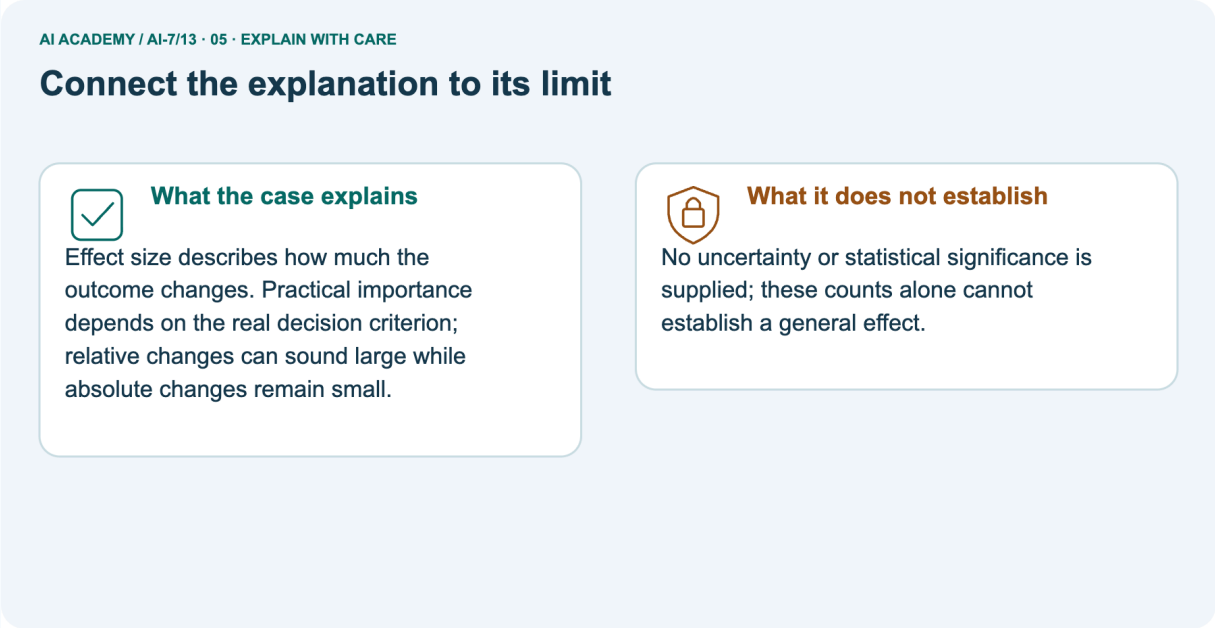
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Five is less than ten, so the observed change misses this threshold. Changed condition: B makes 8 errors in 1000. Expected result: The reduction becomes 12 per 1000 and meets the stated **practical threshold**.

Reason: Twenty minus eight exceeds the required ten fewer errors.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



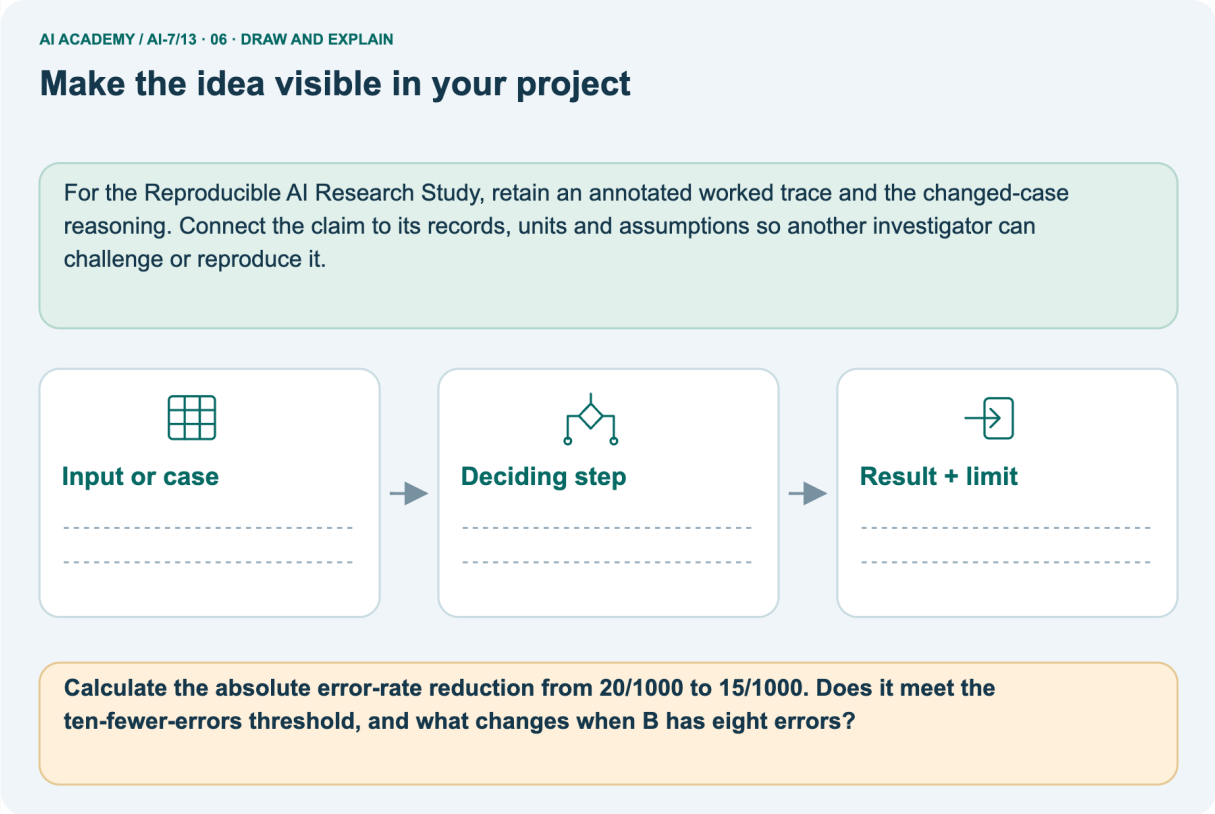
Follow the labelled picture with the matching explanation. Open the vector diagram

Effect size describes how much the outcome changes. Practical importance depends on the real decision criterion; relative changes can sound large while absolute changes remain small.

Limit: No uncertainty or statistical significance is supplied; these counts alone cannot establish a general effect.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Effect size and practical importance: the case

02. Follow the deciding steps

03. Errors on 1000 tasks

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.